NeuroVision: AI-Powered Health Assessment Assitant

Technical Documentation & System Report

1. Executive Summary

NeuroVision is a deep learning framework designed to detect neurodegenerative indicators—specifically Parkinson's Disease (PD)—through the analysis of oculomotor (eye movement) patterns. Unlike traditional diagnostic methods that rely on subjective clinical observation or expensive medical-grade hardware, NeuroVision utilizes a **Bi-Directional LSTM (Long Short-Term Memory)** neural network to analyze temporal gaze sequences. The system includes a webcam-based inference pipeline using **Google MediaPipe**, making it a low-cost, accessible screening tool.

2. Dataset Information

The model is trained on high-precision eye-tracking data obtained from a clinical study involving patients with Parkinson's Disease and a control group.

- **Source Repository:** Borealis Data (University of Waterloo Dataverse)
- **Dataset Name:** Parkinson's Disease Eye Tracking Dataset
- **Persistent ID:** doi:10.5683/SP3/X8YK0S
- **Data Content:**
    - The raw data consists of tabular eye-tracking logs (exported as .txt or .csv).
    - **Participants:** The code differentiates between "Sham" (Control Group/Healthy) and "Participant" (Parkinson's Disease).
    - **Specific File Used:** OH01_April_14_2022 - Raw Data.txt (as seen in the notebook).
    - **Sampling:** The data contains high-frequency "Point of Regard" (gaze) coordinates.

2.1 Feature Selection & Engineering

To ensure the model is compatible with standard webcams (which cannot measure pupil dilation or depth accurately), the training phase deliberately restricts input features to 2D coordinates:

- **Input Features (4 Dimensions):**
    - Point of Regard Left X [px]
    - Point of Regard Left Y [px]
    - Point of Regard Right X [px]
    - Point of Regard Right Y [px]
- **Data Preprocessing Strategy:**
    - **Cleaning:** Removal of NaN values where tracking was lost.
    - **Normalization:** Min-Max normalization is applied per participant to scale pixel values to a 0.0 to 1.0 range. This is crucial for allowing the model to generalize from high-res screen data to arbitrary webcam resolutions.
    - **Windowing:** The continuous time-series data is sliced into sliding windows to capture temporal dynamics.
        - *Window Size:* 300ms (approx. equivalent frames).
        - *Stride:* 100ms (overlap).

3. Model Architecture & Training Parameters

The core of the system is a Recurrent Neural Network (RNN) optimized for sequence classification.

3.1 Architecture Specifications

- **Model Type:** NeuroVisionModel (Custom PyTorch Module)
- **Core Layer: Bi-Directional LSTM**
    - *Input Dimension:* 4 (Lx, Ly, Rx, Ry)
    - *Hidden Size:* 128
    - *Number of Layers:* 2 (Stacked)
    - *Bidirectional:* True (Allows the model to learn from both past and future context within the window).
- **Classification Head (Fully Connected):**
    - Linear Layer (256 $\rightarrow$ 128) $\rightarrow$ ReLU Activation $\rightarrow$ Dropout (0.3) $\rightarrow$ Output Layer (2 Classes).

3.2 Training Configuration

- **Framework:** PyTorch
- **Optimizer:** AdamW (Adam with Weight Decay)
- **Learning Rate:** 1e-3
- **Loss Function:** CrossEntropyLoss (Standard for binary classification).

- **Batch Size:** 32 (Train), 64 (Test).
- **Epochs:** 25
- **Hardware Acceleration:** CUDA (Tesla T4 GPU used in training).

3.3 Performance Results

Based on the training logs provided:

- **Final Training Loss:** Reduced from ~10.56 to ~6.55.
- **Final Test Accuracy: 89.3% (0.893)** reached at Epoch 24.
- **Class Distribution:** The processed dataset contained 607 windows (382 Control / 225 PD), handled via stratified splitting.

4. Competitive Advantages (Comparison with Existing Works)

NeuroVision offers distinct advantages over traditional medical approaches and standard machine learning baselines:

Feature	Traditional Methods / Existing Works	NeuroVision (Proposed Approach)
Hardware Requirement	Requires IR Eye Trackers (e.g., Tobii, EyeLink) costing $10k+.	**Hardware Agnostic:** Works with standard webcams/smartphones via MediaPipe extraction.
Feature Dependency	Often relies on microsaccades or pupil dilation (requires IR sensors).	**Robust Features:** Relies solely on X/Y gaze coordinates, which are extractable from standard video.
Algorithm	Often uses Static Machine Learning (SVM, Random Forest) on aggregate statistics (e.g., "average speed").	**Deep Temporal Learning:** Uses Bi-LSTMs to analyze the *sequence* of movement, capturing subtle tremors and velocity changes over time.
Accessibility	Restricted to clinics/labs.	**Deploys Anywhere:** Can be run as a Python script or mobile app backend.

5. Video Inference Pipeline

The project includes a robust inference engine designed to process standard video files (.mp4) instead of raw text logs.

1. **Face Mesh Extraction:** Uses **Google MediaPipe Vision** to detect 478 face landmarks in real-time.
2. **Iris Isolation:** specifically extracts Landmark indices 468 (Left Iris) and 473 (Right Iris).
3. **Coordinate Transformation:** Converts video frame coordinates to the normalized 0-1 ratio expected by the LSTM.
4. **Padding/Truncation:** Automatically adjusts video length to match the model's input window requirement (300 frames).
5. **Risk Scoring:** outputs a probability score (0.0 to 1.0).
    a. **> 0.75:** High Risk
    b. **> 0.45:** Moderate Risk
    c. **< 0.45:** Low Risk

6. Future Scope of Work: Other Detectable Diseases

While NeuroVision is currently trained on Parkinson's data, the oculomotor system is a window into the brain's neural integrity. The following neurological conditions exhibit distinct eye movement biomarkers and could be detected by retraining the NeuroVision architecture with appropriate datasets:

6.1 Schizophrenia

- **Biomarker:** Smooth Pursuit Eye Movement (SPEM) deficits.
- **Detection Method:** Patients show "saccadic intrusion" (jerky movements) when trying to smoothly track a moving object. The LSTM could easily detect the jagged velocity profile compared to the smooth profile of healthy controls.

6.2 Autism Spectrum Disorder (ASD)

- **Biomarker:** Atypical gaze fixation and social avoidance.
- **Detection Method:** When presented with social stimuli (faces), individuals with ASD often fixate on the mouth or background rather than the eyes. NeuroVision could be adapted to analyze "Heatmap Distribution" features alongside trajectory.

6.3 Multiple Sclerosis (MS)

- **Biomarker:** Internuclear Ophthalmoplegia (INO).
- **Detection Method:** This manifests as a lag in one eye when looking to the side (disconjugate gaze). Since NeuroVision inputs Left and Right eye coordinates separately (Lx, Ly vs Rx, Ry), the model can learn detection patterns where the eyes are not perfectly synchronized.

6.4 Traumatic Brain Injury (TBI) / Concussion

- **Biomarker:** Vergence dysfunction and predictive timing errors.
- **Detection Method:** Analysis of the eyes' ability to converge (cross) when an object moves closer, or the latency (delay) in moving eyes to a new target.

6.5 Attention Deficit Hyperactivity Disorder (ADHD)

- **Biomarker:** Fixation instability and unwanted saccades.
- **Detection Method:** Difficulty maintaining gaze on a fixed point for extended periods. The model would detect high-frequency micro-movements during tasks requiring static attention.